Santhosh bharadwaj H A

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https://santhoshbharadwaj-h-a.github.io/portfolio/

Professional Summary:

- **Data Engineer with 4.9+ years of experience** in designing and implementing high-performance data solutions. Proficient in Python, PySpark, SQL, AWS, and Pandas, with a strong focus on developing scalable data pipelines, optimizing processes, and ensuring data integrity.
- **Skilled in problem-solving** with extensive knowledge of big data technologies and cloud platforms.
- **Successfully built and maintained ETL pipelines** using Python and PySpark to efficiently process and transform large datasets.
- **Enhanced query performance more than 20%** through optimizing SQL queries for faster data retrieval and aggregation.
- **Designed and deployed data warehousing solutions on AWS**, enabling advanced analytics and data-driven decision-making.
- **Engineered seamless data integration workflows**, ensuring efficient data movement across various systems and platforms.
- **Performed thorough data quality assessments**, implementing data cleansing and validation techniques to uphold accuracy and reliability.
- **Collaborated with cross-functional teams** to understand business data needs and deliver customized technical solutions supporting data-driven initiatives.
- **Built RESTful APIs (Django, Flask)** to deliver secure, performant data services for internal tools and external apps.

Professional Experience:

- **Prodapt Solutions Pvt. Ltd.**
  Data Engineer — May 2025 - Present
- **Medilenz Innovation Private Limited**
  Data Engineer — January 2024 – May 2025
- **SMS TechSoft**
  Data Engineer — February 2021 - December 2024

Technical Skill Set:

Programming Languages: Python, SQL, JavaScript
Big Data Technologies: PySpark
Databases: PostgreSQL, MySQL, MariaDB, MongoDB, Mysql, Oracle Amazon Redshift, AWS Aurora, RDS, Athena
Cloud Platforms: Amazon Web Services (AWS)
Data Engineering Tools & Frameworks: Apache Spark, Apache Kafka, AWS Glue, AWS Lambda, AWS EVENTBRIDFE, AWS Step functions, AWS SNS, AWS DMS, AWS Kinesis, Apache Airflow, Django, Restful API's
ETL & Data Processing: Pandas, Redshift, Athena

Containerization & Orchestration: Docker, Kubernetes
Automation & CI/CD: CI/CD practices
Data Visualization Tools: Grafana, Power BI
Agile Practices: Agile methodologies, Task prioritization
Machine Learning Support: Knowledge of AI/ML workflows, Optimization of data pipelines for AI/ML
Development & Version Control: Git, Jupyter Notebook

Education:

- **BE**. in **Information Science** & **Engineering**
  Visvesvaraya Technological University 2017-2021
- **Diploma in computer Application (DCA)**
  Shri Swami Vivekananda institute Graduated: June 2017

Projects:

1.Title	**Large-Scale Customer Data Migration (24 Millon Customer)**
Description	Led the migration of 18M customer records into a unified staging layer and multiple target databases. Built scalable ingestion and transformation pipelines with strong data quality controls. Implemented restart-safe orchestration and auditability for reliable, repeatable runs.
Period	April 2025 – Present
Position	Senior Software Engineer
Responsibilities	• Designed migration blueprint (staging → transform → target) across PostgreSQL, MySQL, MariaDB, MongoDB, and Oracle. • High-throughput S3 ingestion in PySpark; supports schema drift and incremental loads. • AWS Glue transforms to clean, standardize, and dedupe data. • Orchestrated with Glue Workflows & Step Functions; Lambda for checks and alerts. • Built end-to-end reconciliations (source → staging → target): Mini Recon 1/2/3 (row parity, hash totals, key coverage) + Detailed Recon (field-level and business-rule checks) with exception buckets. • Wrote rollback scripts (point-in-time restores, partial replays) with guardrails and audit logs. • Performance optimization: tuned PySpark/Glue (partitioning, broadcast joins, file sizing) to cut runtimes and costs. • Ran performance runs (load tests, backfill drills) and set capacity/throughput baselines. • Led defect triage & fixes with RCA, quick patches, and post-mortems to prevent regressions. • Improved observability: CloudWatch metrics/dashboards, structured logs, alerts, and end-to-end run health.
Technical Skills	**Python, PySpark, SQL, AWS (S3, Glue, Crawlers, Glue Workflows, Lambda, Step Functions), PostgreSQL, MySQL, MariaDB, MongoDB, Oracle, Pandas, Redshift, Git, CI/CD**

2.Title	**GenAI-Powered Medical Record Review Pipeline**
Description	Built an end-to-end pipeline to process unstructured medical records at scale. Implemented OCR → structure → analyze flow, storing normalized outputs in S3 (CSV) and using Amazon Bedrock for clinical summarization and insights.
Period	April 2024 – 2025
Position	Software Engineer
Responsibilities	• Designed ingestion → OCR → normalization → GenAI analysis on AWS (S3-centric; batch & incremental). • Orchestrated AWS Glue Python Shell jobs and Lambdas to trigger Amazon Textract OCR and write structured outputs (CSV/Parquet) to curated S3 prefixes. • Implemented cleaning/normalization with Pandas (field extraction, code mapping, date standardization, schema validation). • Integrated Amazon Bedrock for summarization, entity extraction, and insights with prompt templates, guardrails, and retry policies.
Technical Skills	**Python, Pandas, SQL, AWS (S3, Glue Python Shell, Lambda, Step Functions, CloudWatch), Amazon Textract (OCR), Amazon Bedrock (GenAI), Pandas-to-S3 (CSV/Parquet), Amazon Athena/Redshift (QA/analytics), Git, CI/CD**

3.Title	**Personalized Learning Pathways with Real-Time Feedback.**
Description	develop and maintain an end-to-end data pipeline for a personalized learning platform. This involved extracting, cleaning, and transforming data from various sources, integrating it into a scalable AWS Redshift data warehouse using tools like Apache Airflow and Python to ensure high-quality, real-time data for optimized learning experiences.
Period	February 2021 - February 2024
Position	Data Engineer
Responsibilities	• to extract data from the different data sources • including user interactions, user profiles and feedback. • Build ETL pipelines to integrate and aggregate data from different sources into • a unified data warehouse. Using tools like Apache Airflow, DBT, or custom scripts for this purpose. • Implement data cleaning and transformation processes to ensure data quality. • Use tools like Apache Spark or Pandas for data processing. • Store processed data in a scalable data warehouse (e.g., AWS Redshift) optimized for querying.
Technical Skills	**Python, Pandas, SQL, AWS, Apache Airflow, Redshift**

Declaration

I hereby declare that the information furnished above is complete and true to the best of my knowledge. Place: Bangalore